

		$Q_f = 1500 \text{ J}$
13	B	Loss in kinetic energy of pellet = Gain in thermal energy of pellet $0.5 \times \frac{1}{2}mv^2 = mc\Delta\theta$ Rise in temperature of pellet, $\Delta\theta = \frac{v^2}{4c}$
14	C	From graph $x = 0.30 \text{ m}$ $T = 5.0 \text{ s}$